Formula Sheet

CBSE Class 12 Mathematics • NCERT Chapters 7–13

Integrals • Application of Integrals • Differential Equations • Vector Algebra
Three Dimensional Geometry • Linear Programming • Probability

Blue boxes are NCERT (board) formulas. Purple **JEE extra** boxes go beyond the NCERT book and are not needed for the board exam. Use this sheet with the separate revision notes.

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Chapter 7 Integrals

Basic integrals

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int \frac{1}{x}\,dx = \log|x| + C$$

$$\int e^x\,dx = e^x + C$$

$$\int a^x\,dx = \frac{a^x}{\log a} + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

$$\int \sec^2 x\,dx = \tan x + C$$

$$\int \operatorname{cosec}^2 x\,dx = -\cot x + C$$

$$\int \sec x \tan x\,dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C$$

$$\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + C$$

Integrals of tan, cot, sec, cosec

$$\int \tan x\,dx = \log|\sec x| + C$$

$$\int \cot x\,dx = \log|\sin x| + C$$

$$\int \sec x\,dx = \log|\sec x + \tan x| + C$$

$$\int \operatorname{cosec} x\,dx = \log|\operatorname{cosec} x - \cot x| + C$$

Special integrals (most important, learn all six)

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + C$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + C$$

$$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left|x+\sqrt{x^2-a^2}\right| + C$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left|x+\sqrt{x^2+a^2}\right| + C$$

Integrals of square roots

$$\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right| + C$$

$$\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right| + C$$

$$\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$$

Integration by parts

$$\int u\,v\,dx = u\int v\,dx - \int\left(\frac{du}{dx}\int v\,dx\right)dx$$

Choose u (first function) by **ILATE**:

Inverse trig $\rightarrow$ **L**ogarithmic $\rightarrow$ **A**lgebraic $\rightarrow$ **T**rigonometric $\rightarrow$ **E**xponential

Shortcut: $\int e^x\big[f(x)+f'(x)\big]\,dx = e^x f(x) + C$

Useful trig identities for integration

$$\sin^2 x = \frac{1-\cos 2x}{2}$$
$$\cos^2 x = \frac{1+\cos 2x}{2}$$
$$\sin^3 x = \frac{3\sin x - \sin 3x}{4}$$
$$\cos^3 x = \frac{3\cos x + \cos 3x}{4}$$
$$2\sin A\cos B = \sin(A+B)+\sin(A-B)$$
$$2\cos A\cos B = \cos(A+B)+\cos(A-B)$$
$$2\sin A\sin B = \cos(A-B)-\cos(A+B)$$
$$1+\tan^2 x = \sec^2 x$$

Partial fractions

$$\frac{px+q}{(x-a)(x-b)} \Rightarrow \frac{A}{x-a}+\frac{B}{x-b}$$
$$\frac{px+q}{(x-a)^2} \Rightarrow \frac{A}{x-a}+\frac{B}{(x-a)^2}$$
$$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)} \Rightarrow \frac{A}{x-a}+\frac{B}{x-b}+\frac{C}{x-c}$$
$$\frac{px^2+qx+r}{(x-a)^2(x-b)} \Rightarrow \frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{x-b}$$
$$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)} \Rightarrow \frac{A}{x-a}+\frac{Bx+C}{x^2+bx+c} \quad (x^2+bx+c \text{ cannot be factorised})$$

If degree of numerator $\geq$ degree of denominator, **divide first**.

Definite integrals: Fundamental Theorem and properties

FTC: If $F'(x)=f(x)$, then $\int_a^b f(x)\,dx = F(b)-F(a)$. Also $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$.

P_0 $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$

P_1 $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx, \quad \int_a^a f(x)\,dx = 0$

P_2 $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

P_3 $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

P_4 $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ (the "king" property)

P_5 $\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_0^a f(2a-x)\,dx$

P_6 $\int_0^{2a} f(x)\,dx = 2\int_0^a f(x)\,dx$ if $f(2a-x)=f(x)$; $=0$ if $f(2a-x)=-f(x)$

P_7 $\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$ if f is even; $=0$ if f is odd

Remember: $\int_0^{\pi/2} \log\sin x\,dx = \int_0^{\pi/2} \log\cos x\,dx = -\frac{\pi}{2}\log 2$

Substitution shortcuts

$$\int \frac{f'(x)}{f(x)}\,dx = \log|f(x)|+C$$
$$\int \frac{f'(x)}{\sqrt{f(x)}}\,dx = 2\sqrt{f(x)}+C$$
$$\int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1}+C$$
$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b)+C$$

$px+q = A\dfrac{d}{dx}(ax^2+bx+c)+B$ for $\displaystyle\int \frac{px+q}{ax^2+bx+c}dx$ and $\displaystyle\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$.

JEE extra

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)+C, \qquad \int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)+C$$

Wallis: $\displaystyle\int_0^{\pi/2}\sin^n x\,dx = \int_0^{\pi/2}\cos^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdots\times\begin{cases}\frac{\pi}{2} & n \text{ even}\\ 1 & n \text{ odd}\end{cases}$

Leibniz: $\dfrac{d}{dx}\displaystyle\int_{g(x)}^{h(x)} f(t)\,dt = f(h(x))h'(x) - f(g(x))g'(x)$ **Periodic** (period T): $\displaystyle\int_0^{nT} f = n\int_0^T f$

Limit of a sum: $\displaystyle\int_a^b f(x)\,dx = \lim_{h\to 0} h\big[f(a)+f(a+h)+\cdots+f(a+(n-1)h)\big],\ h=\frac{b-a}{n}$

Substitutions: $\sqrt{a^2-x^2} \to x = a\sin\theta$; $a^2+x^2 \to x = a\tan\theta$; $\sqrt{x^2-a^2} \to x = a\sec\theta$; $\frac{1}{a+b\cos x} \to t = \tan\frac{x}{2}$

$$\int_0^{\pi/2}\frac{f(\sin x)}{f(\sin x)+f(\cos x)}dx = \frac{\pi}{4}, \qquad \int_0^{\pi} x f(\sin x)\,dx = \frac{\pi}{2}\int_0^{\pi} f(\sin x)\,dx$$

Chapter 8 Application of Integrals

Area under a curve

Curve $y = f(x)$, x-axis, $x = a$, $x = b$	$A = \int_a^b y\,dx = \int_a^b f(x)\,dx$
Curve $x = \phi(y)$, y-axis, $y = c$, $y = d$	$A = \int_c^d x\,dy = \int_c^d \phi(y)\,dy$
Curve below the x-axis ($f(x) < 0$)	$A = \left\lvert \int_a^b f(x)\,dx \right\rvert$
Curve crosses the axis at $x = c$	$A = \left\lvert \int_a^c f(x)\,dx \right\rvert + \left\lvert \int_c^b f(x)\,dx \right\rvert$

Standard areas

Circle $x^2 + y^2 = a^2$: πa^2

Quadrant of that circle: $\dfrac{\pi a^2}{4}$

Ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$: πab

Ellipse curve: $y = \dfrac{b}{a}\sqrt{a^2 - x^2}$

Useful: $\displaystyle\int_0^a \sqrt{a^2 - x^2}\,dx = \frac{\pi a^2}{4}$

JEE extra

Between two curves ($f \geq g$ on $[a, b]$): $A = \displaystyle\int_a^b \big[f(x) - g(x)\big]dx$

Horizontal strips: $A = \displaystyle\int_c^d (x_{\text{right}} - x_{\text{left}})\,dy$

$y^2 = 4ax$ and latus rectum $x = a$	$\frac{8a^2}{3}$	$y^2 = 4ax$ and $x^2 = 4by$	$\frac{16ab}{3}$
$y^2 = 4ax$ and $y = mx$	$\frac{8a^2}{3m^3}$	$\lvert x\rvert + \lvert y\rvert = a$	$2a^2$

Chapter 9 Differential Equations

Order, degree, solutions

- **Order** = order of the highest derivative. **Degree** = power of that derivative, defined only if the equation is a polynomial in the derivatives.
- General solution has as many arbitrary constants as the order; a particular solution has none.

Variable separable

$$\frac{dy}{dx} = g(x)\,h(y) \;\Rightarrow\; \int \frac{dy}{h(y)} = \int g(x)\,dx + C$$

Homogeneous equations

$F(\lambda x, \lambda y) = \lambda^n F(x,y)$ (homogeneous of degree n). $\dfrac{dy}{dx} = g\left(\dfrac{y}{x}\right)$ is homogeneous (degree 0).

Put $y = vx$: $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$. For $\dfrac{dx}{dy} = h\left(\dfrac{x}{y}\right)$, put $x = vy$: $\dfrac{dx}{dy} = v + y\dfrac{dv}{dy}$.

Linear equations

$$\frac{dy}{dx} + Py = Q \;(P, Q \text{ in } x) \qquad \text{IF} = e^{\int P\,dx} \qquad y \cdot \text{IF} = \int Q \cdot \text{IF}\,dx + C$$

$$\frac{dx}{dy} + P_1 x = Q_1 \;(P_1, Q_1 \text{ in } y) \qquad \text{IF} = e^{\int P_1\,dy} \qquad x \cdot \text{IF} = \int Q_1 \cdot \text{IF}\,dy + C$$

Common IFs: $e^{\log f} = f$; $P = \frac{1}{x} \to x$; $P = -\frac{1}{x} \to \frac{1}{x}$; $P = \tan x \to \sec x$; $P = \cot x \to \sin x$; $P = \frac{2x}{1+x^2} \to 1 + x^2$.

Growth and decay

$\dfrac{dN}{dt} = kN \;\Rightarrow\; N = N_0 e^{kt}$ (continuous growth at $r\%$: $k = \frac{r}{100}$; doubling time $= \frac{\log 2}{k}$)

JEE extra

$\frac{dy}{dx} = f(ax+by+c)$**:** put $v = ax + by + c$. $\frac{dy}{dx} = \frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$**:** put $x = X + h$, $y = Y + k$ (lines' intersection).

Bernoulli $\frac{dy}{dx} + Py = Qy^n$: put $z = y^{1-n} \;\Rightarrow\; \frac{dz}{dx} + (1-n)Pz = (1-n)Q$.

Exact differentials:

$$x\,dy + y\,dx = d(xy) \qquad \frac{x\,dy - y\,dx}{x^2} = d\left(\frac{y}{x}\right) \qquad \frac{y\,dx - x\,dy}{y^2} = d\left(\frac{x}{y}\right)$$

$$\frac{x\,dy + y\,dx}{xy} = d(\log xy) \qquad \frac{x\,dy - y\,dx}{x^2+y^2} = d\left(\tan^{-1}\frac{y}{x}\right) \qquad \frac{x\,dx + y\,dy}{x^2+y^2} = d\left(\tfrac{1}{2}\log(x^2+y^2)\right)$$

Forming a DE: n arbitrary constants $\Rightarrow$ differentiate n times and eliminate (order n).

Chapter 10 Vector Algebra

Basics

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $|\vec{r}| = \sqrt{x^2+y^2+z^2}$, $\hat{r} = \dfrac{\vec{r}}{|\vec{r}|}$. $\overrightarrow{AB} = \vec{b} - \vec{a}$.

Direction cosines: $l = \dfrac{a}{r}$, $m = \dfrac{b}{r}$, $n = \dfrac{c}{r}$; $l^2+m^2+n^2 = 1$. Collinear: $\vec{b} = \lambda\vec{a}$ ($\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$).

Section formula $(m:n)$: internal $\dfrac{m\vec{b}+n\vec{a}}{m+n}$, external $\dfrac{m\vec{b}-n\vec{a}}{m-n}$, mid-point $\dfrac{\vec{a}+\vec{b}}{2}$.

Scalar (dot) product

$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3$. $\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$, $\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0$.

$\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$; perpendicular $\iff \vec{a}\cdot\vec{b} = 0$; projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$; $\vec{a}\cdot\vec{a} = |\vec{a}|^2$.

$|\vec{a}\pm\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 \pm 2\vec{a}\cdot\vec{b}$; $(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b}) = |\vec{a}|^2 - |\vec{b}|^2$; $|\vec{a}+\vec{b}| \le |\vec{a}| + |\vec{b}|$.

Vector (cross) product

$$\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}. \quad \vec{b}\times\vec{a} = -\vec{a}\times\vec{b};\ \vec{a}\times\vec{a} = \vec{0}.$$

$\hat{i}\times\hat{j} = \hat{k}$, $\hat{j}\times\hat{k} = \hat{i}$, $\hat{k}\times\hat{i} = \hat{j}$; $\hat{i}\times\hat{i} = \vec{0}$. Parallel $\iff \vec{a}\times\vec{b} = \vec{0}$. $\sin\theta = \dfrac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}$.

Unit vector $\perp$ to both: $\pm\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$. Area: parallelogram $|\vec{a}\times\vec{b}|$; triangle $\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$.

JEE extra

Scalar triple product: $[\vec{a}\,\vec{b}\,\vec{c}] = \vec{a}\cdot(\vec{b}\times\vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ = volume of parallelepiped; coplanar $\iff$ 0; tetrahedron volume $\frac{1}{6}[\vec{a}\,\vec{b}\,\vec{c}]$.

Vector triple product: $\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$. **Lagrange:** $|\vec{a}\times\vec{b}|^2 + (\vec{a}\cdot\vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$.

Component of $\vec{a}$ along $\vec{b}$: $\left(\dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}$. Bisector direction: $\hat{a}\pm\hat{b}$. Parallelogram area from diagonals: $\frac{1}{2}|\vec{d_1}\times\vec{d_2}|$.

Chapter 11 Three Dimensional Geometry

Direction cosines and ratios

$l = \cos\alpha,\ m = \cos\beta,\ n = \cos\gamma;\ l^2 + m^2 + n^2 = 1.$ DRs a, b, c: $l = \dfrac{\pm a}{\sqrt{a^2+b^2+c^2}}$, etc.

Line through $P(x_1, y_1, z_1)$, $Q(x_2, y_2, z_2)$: DRs $x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1$; DCs = DRs $\div PQ$.

Equation of a line

	Vector form	**Cartesian form**
Through $\vec{a}$, parallel to $\vec{b}$	$\vec{r} = \vec{a} + \lambda\vec{b}$	$\dfrac{x-x_1}{a} = \dfrac{y-y_1}{b} = \dfrac{z-z_1}{c}$
Through two points $\vec{a}$, $\vec{b}$	$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$	$\dfrac{x-x_1}{x_2-x_1} = \dfrac{y-y_1}{y_2-y_1} = \dfrac{z-z_1}{z_2-z_1}$

Angle between lines; perpendicular and parallel

$$\cos\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{|\vec{b_1}\cdot\vec{b_2}|}{|\vec{b_1}||\vec{b_2}|} \quad \text{(with DCs: } \cos\theta = |l_1l_2 + m_1m_2 + n_1n_2|)$$

Perpendicular: $a_1a_2 + b_1b_2 + c_1c_2 = 0$. Parallel: $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$.

Shortest distance

Skew lines $\vec{r} = \vec{a}_1 + \lambda\vec{b_1}$, $\vec{r} = \vec{a}_2 + \mu\vec{b_2}$: $d = \left|\dfrac{(\vec{b_1}\times\vec{b_2})\cdot(\vec{a_2}-\vec{a_1})}{|\vec{b_1}\times\vec{b_2}|}\right|$

Cartesian: $$d = \frac{\left\|\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}\right\|}{\sqrt{(b_1c_2-b_2c_1)^2 + (c_1a_2-c_2a_1)^2 + (a_1b_2-a_2b_1)^2}}$$

Parallel lines ($\vec{b}$ common): $d = \dfrac{|\vec{b}\times(\vec{a}_2-\vec{a}_1)|}{|\vec{b}|}$

JEE extra: planes and more

Normal form	$\vec{r}\cdot\hat{n}=d;\;\; lx+my+nz=d$
Through a point, normal (a,b,c)	$a(x-x_1)+b(y-y_1)+c(z-z_1)=0$
Intercept form	$\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=1$
Through intersection of two planes	$(a_1x+b_1y+c_1z-d_1)+\lambda(a_2x+b_2y+c_2z-d_2)=0$
Angle between planes	$\cos\theta=\dfrac{\lvert\vec{n}_1\cdot\vec{n}_2\rvert}{\lvert\vec{n}_1\rvert\lvert\vec{n}_2\rvert}$
Angle between line and plane	$\sin\phi=\dfrac{\lvert\vec{b}\cdot\vec{n}\rvert}{\lvert\vec{b}\rvert\lvert\vec{n}\rvert}$
Distance of point from plane	$\dfrac{\lvert ax_1+by_1+cz_1-d\rvert}{\sqrt{a^2+b^2+c^2}}$; parallel planes $\dfrac{\lvert d_1-d_2\rvert}{\sqrt{a^2+b^2+c^2}}$
Two lines coplanar	$(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)=0$
Foot F of perpendicular from P	$\overrightarrow{PF}\cdot\vec{b}=0$ (F general point); image $=2F-P$

Chapter 12 Linear Programming

Standard form

Optimise (maximise or minimise) $Z = ax + by$ subject to linear constraints $a_i x + b_i y \ (\leq, \geq) \ c_i$ and $x \geq 0$, $y \geq 0$.
Feasible region: common region of all constraints (always convex).

Corner point method

- **Theorem 1:** if an optimal value exists, it occurs at a corner point of the feasible region.
- **Theorem 2:** if the region is **bounded**, Z has both a maximum and a minimum, each at a corner.
- **Unbounded:** if m is the smallest corner value, m is the minimum only if $Z < m$ has no common point with the region. If M is the largest, it is the maximum only if $Z > M$ has no common point.
- **Multiple optima:** if two corners give the same optimal value, every point on the segment joining them does too.
- **No feasible region** $\Rightarrow$ no optimal solution.

JEE extra

Corner of $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$: $x = \dfrac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1}$, $y = \dfrac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$.

Iso-profit line $ax + by = k$: slide it parallel; the last corner touched gives the maximum (first gives the minimum). If it is parallel to an edge of the region, that whole edge is optimal.

Chapter 13 Probability

Basic rules

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $P(A') = 1 - P(A)$; $P(A' \cap B') = 1 - P(A \cup B)$.

$P(A \cap B') = P(A) - P(A \cap B)$; $P(\text{exactly one of } A, B) = P(A) + P(B) - 2P(A \cap B)$.

Conditional probability

$$P(E|F) = \frac{P(E \cap F)}{P(F)}, \quad P(F) \neq 0; \quad 0 \leq P(E|F) \leq 1.$$

$P(S|F) = P(F|F) = 1$; $P(E'|F) = 1 - P(E|F)$; $P\big((A \cup B)|F\big) = P(A|F) + P(B|F) - P\big((A \cap B)|F\big)$.

Multiplication theorem and independence

$P(E \cap F) = P(E)\,P(F|E) = P(F)\,P(E|F)$; $P(E \cap F \cap G) = P(E)\,P(F|E)\,P(G|E \cap F)$.

Independent: $P(E \cap F) = P(E)P(F) \iff P(E|F) = P(E)$. Then E' and F, E and F', E' and F' are also independent.

Three events are mutually independent if every pair, and all three together, satisfy the product rule.

$P(\text{at least one of } A, B) = 1 - P(A')P(B')$ (independent). Mutually exclusive: $P(A \cap B) = 0$.

Total probability and Bayes' theorem

$E_1, \ldots, E_n$ a partition of S (disjoint, union S, each $P(E_i) > 0$):

Total probability: $P(A) = \sum_{j=1}^{n} P(E_j)\,P(A|E_j)$ Bayes: $P(E_i|A) = \dfrac{P(E_i)\,P(A|E_i)}{\sum_{j=1}^{n} P(E_j)\,P(A|E_j)}$

JEE extra

Random variable: $E(X) = \sum x_i p_i$; $\text{Var}(X) = E(X^2) - [E(X)]^2$; $\sum p_i = 1$.

Binomial: $P(X = r) = \binom{n}{r} p^r q^{n-r}$, $q = 1 - p$; mean np, variance npq. $P(\text{at least one success}) = 1 - q^n$.

Alternate turns, A first: $P(A \text{ wins}) = \dfrac{p}{1 - q^2}$. **3 independent events:** $P(\text{at least one}) = 1 - P(A')P(B')P(C')$.

$P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C)$.